

The University Of Azad Jammu & Kashmir

Lab Task # 4

Database Systems (CS-2204)

Instructor's Name

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Database Normalization

(Schema + MySQL Implementation)

1. First Normal Form (1NF)

A relation is in 1NF if all attributes contain atomic values and there are no repeating groups.

Implementation Notes: * The original hospital data was flattened into a single table where every cell contains a single value.

- **Primary Key:** VisitID.
- **Observation:** While atomic, this table has high redundancy; patient and doctor details are repeated for every visit.

Server: 127.0.0.1 » Database: hospital_management » Table: hospital_1nf

total, Query took 0.0003 seconds.)

hospital_1nf`

[Edit] [Explain SQL] [Create PHP code] [Refresh]

Number of rows: 25 Filter rows: Search this table Sort by key: None

	VisitID	VisitDate	PatientID	PatientName	PatientPhone	DoctorID	DoctorName	Specialty	DeptName	DeptHead	Diagnosis	Fee
Delete	V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Hypertension	2500
Delete	V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Eczema	2000
Delete	V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Allergy	2000
Delete	V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Arrhythmia	3000

With selected: Edit Copy Delete Export

Number of rows: 25 Filter rows: Search this table Sort by key: None

Console

Server: 127.0.0.1 » Database: hospital_management » Table: hospital_1nf

Showing rows 0 - 3 (4 total, Query took 0.0007 seconds.)

SELECT * FROM `hospital\_1nf`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	VisitID	VisitDate	PatientID	PatientName	PatientPhone	DoctorID	DoctorName	Specialty	DeptName	DeptHead
☐ Edit Copy Delete	V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq
☐ Edit Copy Delete	V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma
☐ Edit Copy Delete	V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma
☐ Edit Copy Delete	V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq

Check all | With selected: Edit Copy Delete Export

2. Second Normal Form (2NF)

A relation is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on the primary key (no partial dependencies).

Justification of Changes:

- **Removed Partial Dependency 1:** PatientName and PatientPhone depend only on PatientID, not the VisitID.
- **Removed Partial Dependency 2:** DoctorName, Specialty, and DeptName depend only on DoctorID, not the VisitID.
- **Action:** These were extracted into separate Patient and Doctor tables.

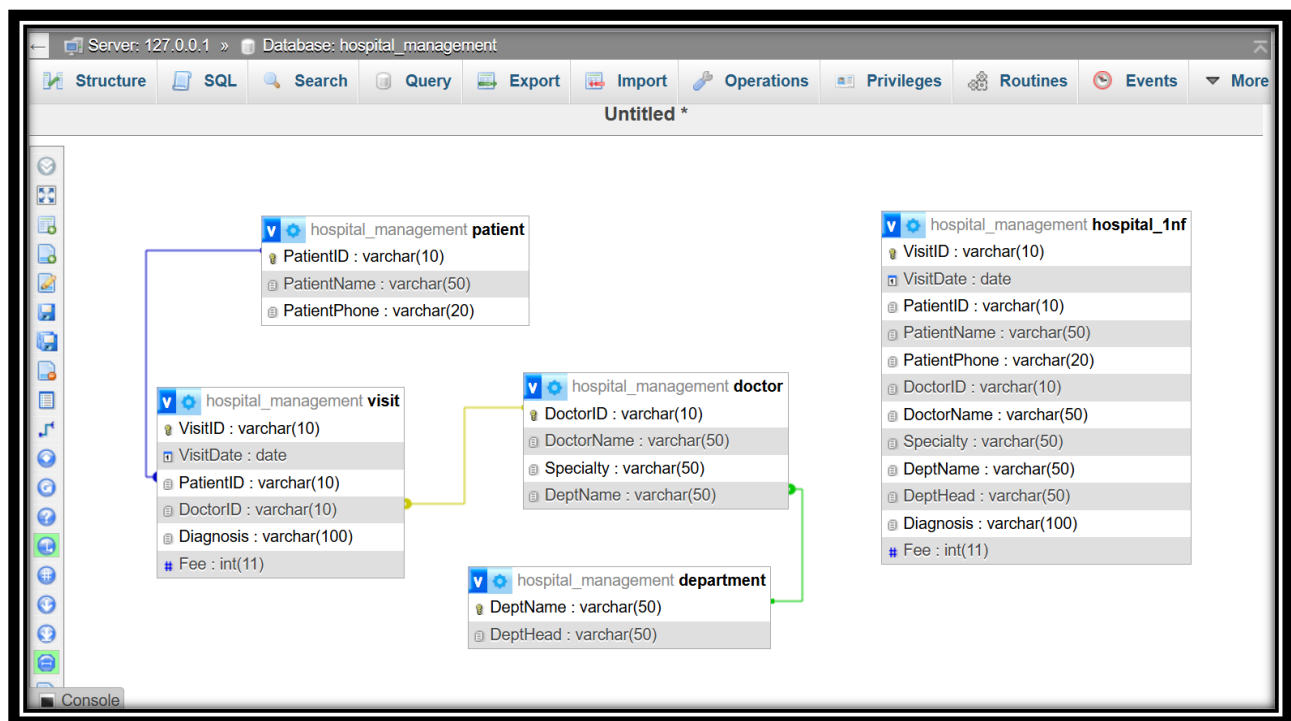
3. Third Normal Form (3NF) - Final Schema

Definition: A relation is in 3NF if it is in 2NF and has no transitive dependencies (no non-prime attribute depends on another non-prime attribute).

Justification of Changes:

- **Removed Transitive Dependency:** In the Doctor table, DeptHead was dependent on DeptName, which was dependent on DoctorID.
- **Action:** A separate Department table was created to hold department-specific details, linked to the Doctor table via DeptName.

Designer View:





4. Verification of Data Integrity

To ensure the normalized schema remains consistent with the original requirements, a JOIN query was executed to reconstruct the original visit report. This confirms that no data was lost during the decomposition process.

Server: 127.0.0.1 » Database: hospital_management

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 3 (4 total, Query took 0.0004 seconds.)

```
SELECT v.VisitID, v.VisitDate, p.PatientID, p.PatientName, p.PatientPhone, d.DoctorID, d.DoctorName, d.Specialty, dep.DeptName,
dep.DeptHead, v.Diagnosis, v.Fee FROM visit v JOIN patient p ON v.PatientID = p.PatientID JOIN doctor d ON v.DoctorID = d.DoctorID JOIN
department dep ON d.DeptName = dep.DeptName;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

VisitID	VisitDate	PatientID	PatientName	PatientPhone	DoctorID	DoctorName	Specialty	DeptName	DeptHead	Diagnosis	Fee
V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Hypertension	2500
V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Eczema	2000
V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Allergy	2000
V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Arrhythmia	3000

Elimination of Anomalies

The final 3NF design prevents the following data integrity issues:

- **Insertion Anomaly:** We can now register a new doctor or department in the system even if they haven't seen a patient yet.
- **Update Anomaly:** If a Department Head changes, we only need to update one row in the Department table instead of hundreds of individual visit records.
- **Deletion Anomaly:** Deleting a specific visit record no longer results in the accidental loss of a patient's contact information or a doctor's specialty details.